

Lecture Notes VII

Course: Functional Analysis

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Topic: Lebesgue Integration on $\mathbb{R}^n$

Proof 1.57 (Cartesian product of opens is open). Let G_n, G_m be open in $\mathbb{R}^n, \mathbb{R}^m$, respectively, then given any point $x \in G_n \times G_m$ we have that $x = (p, q)$ with $p \in G_n, q \in G_m$ so there exist open cubes $B_\delta^\infty(p)$ and $B_{\delta'}^\infty(q)$ contained in G_n, G_m , respectively, so that taking $\varepsilon = \inf\{\delta, \delta'\}$ we have that $B_\varepsilon^\infty(p) \subseteq G_n, B_\varepsilon^\infty(q) \subseteq G_m$ so clearly $B_\varepsilon^\infty(p) \times B_\varepsilon^\infty(q) = B_\varepsilon^\infty(x) \subseteq G_n \times G_m$. We therefore conclude that for any point in $G_n \times G_m$ there exists an open cube contained in $G_n \times G_m$, i.e., $G_n \times G_m$ is open.

Proof 1.58 (Rational balls form a basis for the Euclidean topology of $\mathbb{R}^n$). ($\Rightarrow$) Let G be any open in $\mathbb{R}^n$, then for any $g \in G$ there is some $\delta > 0$ such that $B_\delta^\infty(g) \subseteq G$ and there is some $r \in \mathbb{Q}$ with $0 < r < \delta$ so clearly $B_r^\infty(g) \subseteq B_\delta^\infty(g) \subseteq G$ and thus for each $g \in G$ choose such a $B_r^\infty(g)$. We claim that

$$G = \bigcup_{g \in G} B_r^\infty(g).$$

The inclusion $G \subseteq \bigcup_{g \in G} B_r^\infty(g)$ is just the definition, for the reverse inclusion let $p \in \bigcup_{g \in G} B_r^\infty(g)$, then $p \in B_r^\infty(g)$ for some $r \in \mathbb{Q}$ and some $g \in G$, but $B_r^\infty(g)$ is also such that $B_r^\infty(g) \subseteq G$ by its definition, so $p \in G$ and the proof is done.

By the definition of σ -algebra (particularly, closure under countable unions) we have that $\mathcal{B}^n$ is the smallest sigma algebra containing the opens in $\mathbb{R}^n$, but since opens are exactly those sets that can be written as countable unions of open cubes, we have that $\mathcal{B}^n$ is also the smallest σ -algebra containing all the open cubes.

Proof 1.61 (The product σ -algebra of Borel σ -algebras is a Borel σ -algebra). This is a corollary (via induction) of Theorem 1.18.

Proof 1.62 (Measure of a dilation is the dilation of the measure). Define the collection

$$\mathcal{E} = \{E \in \mathcal{B}^n : tE \in \mathcal{B}^n\} \subseteq \mathcal{B}^n$$

and let $G \in \mathcal{E}$. Then for $\varrho \in tG$ (for which we have that $\varrho = tg$ for some $g \in G$ and since G is open, we also have some $\varepsilon > 0$ s.t. $B_\varepsilon(g) \subseteq G$) we claim that $B_{t\varepsilon}(\varrho) \subseteq tG$. Given any $p \in B_{t\varepsilon}(\varrho)$ we have $\|p - \varrho\|_\infty < t\varepsilon$ so $\|p/t - \varrho/t\| = \|p/t - g\| < \varepsilon$ which means that $p/t \in B_\varepsilon(g) \subseteq G$ whence $p/t \in G$ and thus $p \in tG$. We therefore conclude that if G is open, then so is tG so $\mathcal{E}$ contains every open subset of $\mathbb{R}^n$. Further, since

$$t(\mathbb{R}^n \setminus E) = \mathbb{R}^n \setminus tE \quad \text{and} \quad t\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \bigcup_{n \in \mathbb{N}} tE_n,$$

then $\mathcal{E}$ is a σ -algebra and thus $\mathcal{B}^n \subseteq \mathcal{E}$ so by definition of $\mathcal{E}$ this in turn implies that $\mathcal{E} = \mathcal{B}^n$.

To prove the measure property first notice that for $n = 1$ we have that (recall the definition of the Lebesgue measure as the restriction of the Lebesgue outer measure) is just homogeneity of the Lebesgue outer measure. For $n > 1$ suppose the result holds for $n - 1$, then taking any $A \in \mathcal{B}^{n-1}, B \in \mathcal{B}$ we have $\lambda^n(t(A \times B)) = \lambda((tA) \times (tB)) = \lambda^{n-1}(tA)\lambda(tB)$ (see Example 1.52) $= t^{n-1}\lambda^{n-1}(A)t\lambda(B)$ (by the induction hypothesis) $= t\lambda^n(A \times B)$ (*). Next, for any $m \in \mathbb{N}$ let

$$\mathcal{E}_m = \{E \in \mathcal{B}^n : E \subseteq B_m^\infty(0) \text{ and } \lambda^n(tE) = t^n \lambda^n(E)\}.$$

Given the previous result (*) and given that (see Theorem 1.47) any measurable set is a finite disjoint union of Borel measurable sets, we have that any finite union of Borel measurable rectangles contained in $B_m^\infty(0)$ is again in $\mathcal{E}_m$ and for any increasing sequence (E_k) in $\mathcal{E}_m$ we have that

$$\lambda^n\left(\bigcup_{k \in \mathbb{N}} tE_k\right) = \lim \lambda^n(tE_k) = t^n \lim \lambda^n(E_k) = t^n \lambda^n\left(\bigcup_{k \in \mathbb{N}} E_k\right)$$

(we used continuity from below) and thus $\mathcal{E}_m$ is closed under countable increasing unions and similarly for countable decreasing intersections ($\lambda^n(E_1)$ is finite because it is contained in $B_m^\infty(0)$ given a sequence (E_n) in $\mathcal{E}_m$) so $\mathcal{E}_m$ is a monotone class. We can reason similarly as in the case of $\mathcal{E}$ to deduce that $\mathcal{E}_m$ contains all the Borel measurable rectangles that are themselves contained in $B_m^\infty(0)$. We now take our results all together: $\mathcal{E}_m \supseteq \mathcal{A}$ with $\mathcal{A}$ the collection of finite unions of Borel measurable rectangles contained in $B_m^\infty(0)$ (which is an algebra by Theorem 1.47) and is also a monotone class on $B_m^\infty(0)$ so $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{E}_m$ but by the monotone class theorem this in turn yields $\sigma(\mathcal{A}) \subseteq \mathcal{E}_m$ with $\sigma(\mathcal{A})$ the smallest sigma algebra containing the Borel measurable rectangles contained in $B_m^\infty(0)$ and the definition of $\mathcal{E}_m$ directly implies that $\mathcal{E}_m \subseteq \sigma(\mathcal{A})$ and hence $\mathcal{E}_m = \sigma(\mathcal{A})$. We thus conclude that $\mathcal{E}_m$ consists on the Borel measurable sets that are contained in $B_m^\infty(0)$.

Thus $\lambda^n(tE) = t^n \lambda^n(E)$ for any Borel measurable E contained in $B_m^\infty(0)$ (but $m \in \mathbb{N}$ was arbitrary, so this is not an important restriction as we shall prove next). Suppose $E \in \mathcal{B}^n$, then it is clear that $E = \mathbb{R}^n \cap E$ but $\mathbb{R}^n = \bigcup_{m \in \mathbb{N}} B_m^\infty(0)$ so

$$\lambda^n(tE) = \lambda^n \left(t \left(\bigcup_{m \in \mathbb{N}} (E \cap B_m^\infty(0)) \right) \right) = \lambda^n \left(\bigcup_{m \in \mathbb{N}} t(E \cap B_m^\infty(0)) \right) = \lim \lambda^n(t(E \cap B_m^\infty(0)))$$

but here, notice that since E is Borel measurable, then $E \cap B_m^\infty(0)$ is Borel measurable and contained in $B_m^\infty(0)$ so the previous result implies that

$$\lim \lambda^n(t(E \cap B_m^\infty(0))) = t^n \lim \lambda^n(E \cap B_m^\infty(0)) = t^n \lambda^n(E).$$

Completing the proof.

Proof 1.63 (Formula for the hypervolume of the n -ball). For $n = 1$ this is just the measure of the interval $(-1, 1)$ which is just 2 and hence the formula is correct and similarly for $n = 0$ adopting the convention that $\mathbb{R}^0$ is a single point and $\lambda^0(\{0\}) = 1$. So we assume that $n > 1$ and use strong induction (assume that for any $k \leq n$, the assertion holds true) so

$$\lambda^n(\mathbb{B}^n) = \int_{\mathbb{R}^2} \int_{\mathbb{R}^{n-2}} \chi_{\mathbb{B}^n}(x, y)$$

where any integral is understood to be taken with respect to the appropriate Lebesgue measure. Fixing $x = (x_1, x_2)$ for the first integration we see that